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Nicholas G. Hall

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Scheduling Problems With Generalized Due Dates

NICHOLAS G. HALL

The Ohio State University
301 Hagerty Hall
1775 College Road
Columbus, OH 43210-1399

Abstract: We define a new class of scheduling problems, for which due dates are specified in terms of the position in which a job appears in an ordered sequence, rather than by that job's identity. This problem class appears rich both in applications and in theoretical issues. For many traditional scheduling problems there exists an equivalent problem within the new class. We examine the algorithmic complexity of several of these problems, and attempt to provide an efficient solution technique, or alternatively to demonstrate that such a technique is highly unlikely to exist. In two cases, the complexity of the problem being discussed remains open.

■ This paper provides a new perspective on the traditional view of due dates in the context of scheduling problems. An informal statement of the traditional due date problem might be: "Given jobs A , B and C , schedule them so that job A is completed by time t_A , job B by time t_B and job C by time t_C ." We discuss below several applications where a more appropriate statement would be: "Given jobs A , B and C , schedule them so that one job is completed by time t_1 , a second by time t_2 , and the third job by time t_3 ." We describe due dates which are defined in this way as *generalized due dates*, while referring to due dates defined in the traditional (job-specific) manner as *specific due dates*.

Scheduling problems with generalized due dates occur naturally where a single customer places no precedence relationships or priority orderings on the jobs being performed. A public utility company may face a contract to provide a service to 50 new cities, to include 20 within 6 months, a further 15 within a year and all 50 within 2 years. An airline possessing a license to fly certain routes may face a similar problem. Another application of this problem class arises in survey design, as used in market research studies. An appropriate survey schedule could involve performing surveys in perhaps 20 cities out of 30 within 3 months, without regard to which cities were surveyed. Costs associated with lateness may include overtime and penalties arising from delays in completing the whole project.

The new class of scheduling problems may be useful in a flexible manufacturing system (FMS) environment, of the kind described by Stecké & Solberg [10]. The aspect of flexibility, discussed by Browne, Dubois, Rathmill, Sethi, and Stecké [2], arises in reductions in machine setup costs and efficient change of component mix, making the order of job arrival less important, a situation which lends itself to the generalized due date model.

GENERAL PROBLEM DESCRIPTION

We consider the basic n job machine scheduling problem, the assumptions of which have been described by Baker [1], and define the following notation:

- p_j = processing time of job j
- r_j = ready time of job j
- w_j = weight or value of job j
- C_j = completion time of job j
- $C_{[j]}$ = completion time of the j th job to be processed
- $g_{[j]}$ = due date of the j th job to be processed
- $T_{[j]}$ = $\max [0, C_{[j]} - g_{[j]}]$, the tardiness of the j th job to be processed
- $U_{[j]}$ = 1 if $C_{[j]} > g_{[j]}$
= 0 otherwise.

No specific assumptions are made about the schedule $g_{[j]}$ of generalized due dates, other than that it is a non-

decreasing function of its argument j . Thus $g_{[i]} \leq g_{[k]}$ for all $1 \leq i < k \leq n$. Since this problem definition is very general, many of the standard problems in the scheduling literature (see, for example, Baker [1], Conway, Maxwell and Miller [3], Lenstra, Rinnooy Kan, and Brucker [7]) admit an alternative formulation using the generalized due date concept. We will examine the algorithmic complexity of a number of these problems, and make comparisons with their complexity under a specific due date definition. An introduction to algorithmic complexity can be found in the works of Garey & Johnson [4] and Papadimitriou & Steiglitz [9]. Surveys of results in this area can be found in Lenstra, Rinnooy Kan, and Brucker [7] and Lawler, Lenstra, and Rinnooy Kan [6]. We follow the notation of Lenstra, Rinnooy Kan, and Brucker [7] in describing a scheduling problem by $u | v | x | y$, where u is the number of jobs, v is the number of machines, x describes the type of processing shop and any additional constraints, and y describes the optimality criterion.

Problem #1: $n | 1 | \Sigma T_{[j]}$

This problem requires the minimization of total tardiness, with each job having a processing time. The specific due date version of this problem (where each job also has a due date) is open as to complexity.

Theorem 1

The generalized due date version of Problem #1 is optimized by a semiactive SPT (shortest processing time) ordering of the jobs.

Theorem 1 can be proved by a straightforward adjacent pairwise interchange argument. An SPT ordering can be found in $O(n \log n)$ time.

Problem #2: $n | 1 | \Sigma U_j$

This problem requires the minimization of the number of late jobs, where each job has a processing time and a due date. The specific due date version of this problem was solved using an efficient algorithm by Moore [8].

Theorem 2

The generalized due date version of Problem #2 is optimized by a semiactive SPT ordering of the jobs.

The proof follows that for Theorem 1.

Problem #3: $n | 1 | \Sigma w_j U_j$

This problem requires the minimization of the total weight of late jobs, or equivalently the maximization of the value of jobs which are on time. The specific due date version of this problem was shown by Karp [5] to be NP-complete.

Theorem 3

The recognition version of Problem #3 is NP-complete.

Proof

The recognition version of Problem #3 is, "Does there

exist a feasible schedule which has a total weight of late jobs less than or equal to some number R ?" Given a 'yes' answer and a schedule, it is easy to check that the schedule is feasible and compute the total weight of late jobs, in polynomial time. This means that the problem is a member of the class NP (see Garey & Johnson [4]). Next, specialize the problem by setting $g_{[j]} = K$, $j = 1, \dots, n$. The recognition problem is then equivalent to the problem of deciding whether there exists a solution of value less than or equal to R to the problem

$$\text{MAX} \sum_{j=1}^n w_j X_j$$

$$\text{s.t.} \sum_{j=1}^n p_j X_j \leq K$$

$$X_j = 0 \text{ or } 1, j = 1, \dots, n.$$

But the latter problem is the recognition version of (zero-one) Knapsack, an NP-complete problem.

Problem #4: $n | 1 | r_j \geq 0 | \Sigma T_{[j]}$

Here we aim to minimize total tardiness, as in Problem #3, but this time in the presence of nonzero ready times. The specific due dates version of this problem was shown to be NP-complete by Lenstra, Rinnooy Kan, and Brucker [7].

Theorem 4

The recognition version of Problem #4 is NP-complete.

Proof

The proof of NP-membership for the recognition version of this problem follows that for Problem #3. We specialize the problem by setting $g_{[j]} = 0$, $j = 1, \dots, n$. Then the problem is equivalent to the recognition version of $n | 1 | r_j \geq 0 | \Sigma C_{[j]}$ and the latter problem was shown by Lenstra, Rinnooy Kan, and Brucker [7] to be NP-complete.

Problem #5: $n | 2 | \text{MAX}_{1 \leq j \leq n} \{C_{[j]} - g_{[j]}\}$

This problem requires the minimization of the maximum lateness among all the jobs, using two identical parallel processors. This problem defined with specific due dates is NP-complete, as again demonstrated by Lenstra, Rinnooy Kan, and Brucker [7].

Theorem 5

The recognition version of Problem #5 is NP-complete.

Proof

As before, the recognition version of the problem is in NP. Next we follow Papadimitriou & Steiglitz [9, p. 346] in making the very general assumption that the cost of the optimal solution is an integer with logarithm bounded by a polynomial function of the size of the in-

put. This being the case, we can answer the optimization Problem #5 by answering the following equivalent recognition version "Does there exist a schedule in which no job is later than some integer K , relative to its due date?" Now we specialize the problem by setting $g_{[1]}$

$$= \frac{1}{2} \cdot \sum_{j=1}^n p_j + K, j = 1, \dots, n, \text{ and we will answer}$$

the recognition problem with a "yes" if and only if all the jobs can be scheduled on time on two processors. Consider the special case $K = 0$ and we have an NP-complete problem known as partition. Partition can be stated as "Given numbers $X_1, \dots, X_n$ and a set $S = \{1, \dots, n\}$, do there exist subsets $S_1 \in S, S_2 \in S$, such that

$$S_1 \cup S_2 = S, S_1 \cap S_2 = \emptyset, \text{ and } \sum_{j \in S_1} X_j = \sum_{j \in S_2} X_j?"$$

Since this problem can be decided by whether or not a schedule with no late jobs is possible, the recognition version of Problem #5 is NP-complete.

Problem #6: $n | 2 | \Sigma U_j$

The minimization of the number of late jobs using two identical parallel processors is required here. The version of this problem with specific due dates was shown by Lenstra, Rinnooy Kan, and Brucker [7] to be NP-complete.

Theorem 6

The recognition version of Problem #6 is NP-complete. The proof follows that for Problem #5.

Two generalized due date problems remain open, minimizing maximum lateness with ready times ($n | 1 | r_j \geq 0 | \text{MAX}_{1 \leq j \leq n} \{C_{[1]} - g_{[1]}\}$) and minimizing

weighted tardiness ($n | 1 | \Sigma w_{[1]} T_{[1]}$). The specific due date case for each is NP-complete [7].

Conclusions

This paper has introduced a new class of scheduling problems, many of which are closely related to well-studied problems in the literature. Potential applications exist in survey design and a variety of scheduling contexts, including FMS. Although the new problems are intuitively easier to solve than corresponding problems with the established definition of due dates, there appears to be little reduction in complexity for many of these new problems. The new problem class generates a variety of

open questions, of which we give two examples, in the area of problem complexity.

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Nicholas G. Hall is an Instructor in the Faculty of Management Sciences, The Ohio State University, 1775 College Road, Columbus, Ohio 43210-1399. He holds a B.A., M.A. in Economics from the University of Cambridge, is a Chartered Accountant in England, and is currently completing his Ph.D. in Management Science at the University of California, Berkeley. His research interests are in the analysis of heuristics, machine scheduling, inventory theory and network flow problems. He has published in a variety of journals including *Naval Research Logistics Quarterly*, *European Journal of Operational Research*, *Discrete Applied Mathematics* and *International Journal of Production Research*.